

Towards TRANSIENT ELECTRONICS And The Need For It



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Electronic devices are built to last. But now their lifespans can be controlled, courtesy burgeoning research into transient electronics, or devices that are meant to serve a specific function before completely dissolving into their environment over a predetermined span of weeks, months or even years. Moreover, with a degrading environment where electronic waste also plays a major role, the need for such electronic devices that can be disposed of when we want increases.

The generation of heaps of electronic waste around us is compelling scientists to look towards the development of devices that will dissolve either on command or with time.

The technology that can break electronics down using a specific environmental trigger would allow metals and other non-biodegradable elements to dissolve down to their molecular elements for recycling. Self-destructing electronic devices is a step towards greatly reducing electronic waste and boosting sustainability in the use of electronic devices in our daily lives.

ble electronics, environmental monitoring applications and so on.

Millions of patients have benefited from the innovative development of medical electronic devices such as pacemakers or medicine-dispensing agents implanted in the body for either diagnostic or therapeutic benefits. While many such devices have revolutionised modern medicine, these often outlive their purpose in the body and require surgical removal to avoid complication.

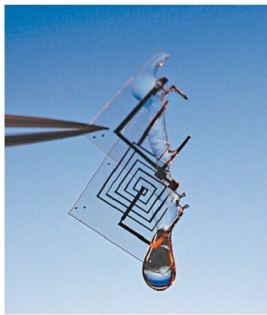
But rather than surgery, what if these could simply disappear? That is the concept behind transient electronics. Transient electronics offers robust performance compared to current devices but fully resorbs into the environment after appropriate time—ranging from minutes to years, depending on the application.

To explore these possibilities, scientists have been working with an array of natural materials to test how well these perform in electronic devices and whether these would cause side effects or damage when implanted. One day it may transform the consumer electronics industry, but researchers are currently more interested in the devices' potential for medical care through implantable sensors or drug dispensers, as well as military exploits and environmental monitoring. Research into such devices has made rapid progress; preliminary versions of vanishing electronics might be available within the next few years.

Applications

Medical implants that are only needed for a few weeks could just disappear thereafter, without requiring extra surgery to remove these from the body. And, no one would have to retrieve dozens of transient water-quality sensors from a river undergoing water-quality monitoring. The electronics would dissolve

Fig. 1: Transient electronics can dissolve entirely in water or other liquids over a period of time ranging from minutes to weeks, a capability that offers a wide range of medical, environmental and other uses (Image courtesy: University of Illinois/Beckman Institute)



Potential of transient electronics

From medicine to military, transient electronics has many potential applications, specially where sensitive data is used. Applications include military electronic equipment, credit cards, passports and any application in which the electronic device is to be used for a defined period of time, for example, bio-electronics, implanta-